

ADNAN PATEL

Product Engineer | Software Development | AI/ML | Python | Cybersecurity

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PROFESSIONAL SUMMARY

Product Engineer with professional experience in enterprise software and observability, working on Go-based plugin development, AWS integrations, REST API troubleshooting, and log/telemetry analysis for a large-scale monitoring platform. Skilled in Python, backend/application development, structured debugging, and cross-functional collaboration across Product, QA, and Engineering teams. Complements this professional background with hands-on experience in AI/ML and cybersecurity, including machine learning, intrusion detection, reconnaissance, vulnerability assessment, API security testing, and attack simulation. Comfortable working across software development, API integration, AI/ML, and technical troubleshooting.

TECHNICAL SKILLS

AI / Machine Learning (Technical Experience): TensorFlow, Keras, Scikit-learn, XGBoost, Pandas, NumPy, CNN, Prompt Engineering, Prompt Design, Prompt Evaluation, Model Training, Data Preprocessing, Model Evaluation

Programming Languages: Python, Golang, PHP, JavaScript, Bash

Backend & API Development: Flask, REST APIs, API Integrations, Postman, cURL, JSON, HTTP, API Debugging

Cloud & Infrastructure: AWS (EC2, CloudWatch, ECS, EKS, ECR, IAM, VPC, AWS Backup), Distributed Systems, Containerized Workloads

Product Engineering & Observability: Log Analysis, Telemetry Monitoring, Plugin Development (Golang), Root Cause Analysis, Incident Management, Metrics Dashboards, Alerting, Product Validation

Software Engineering: SDLC, Agile/Scrum, Technical Documentation, Cross-functional Collaboration

Cybersecurity & Networking: Reconnaissance, Vulnerability Assessment, API Security Testing, Attack Simulation, TCP/IP, Nmap, Recon-ng, Traffic Analysis, Endpoint Diagnostics, Linux Troubleshooting

Tools & Platforms: Git, GitHub, Cursor AI, Jupyter Notebook, JupyterLab, VS Code, IntelliJ IDEA, GoLand, Linux CLI, Google Colab, Figma

EDUCATION

Bachelor of Technology: Computer Engineering

2026

P P Savani University, Surat

EXPERIENCE

Product Engineer (Enterprise Observability Platform)

Sept 2025 – May 2026

Motadata (Mindarray Systems Pvt. Ltd.) – Ahmedabad

- Developed and enhanced Go-based monitoring plugins and integrations for AWS, HPE MSA Storage, and Harmonica within Motadata's ObserveOps enterprise observability platform.
- Analyzed service telemetry, logs, and monitoring data to improve data visualization, observability, and actionable insights for end users.
- Performed competitive analysis of enterprise monitoring platforms to evaluate capabilities, identify feature gaps, and support new product feature planning and enhancement initiatives.
- Collaborated with Product, QA, and Backend Engineering teams to translate customer requirements into reliable monitoring solutions and validate new features.
- Authored technical documentation and workflow diagrams to improve troubleshooting efficiency and knowledge sharing across engineering teams.
- Investigated REST API integration issues, schema mismatches, and runtime failures through structured debugging and root cause analysis.
- Authored technical documentation, plugin workflows, and validation reports to improve troubleshooting efficiency and knowledge sharing.
- Supported AWS cloud monitoring environments, product validation, and enterprise deployments while participating in Agile development practices.

PROJECTS

Intrusion Detection & Blocking System

- Developed an end-to-end Machine Learning-based Intrusion Detection System (IDS) by training and evaluating a Random Forest classifier on the NSL-KDD dataset to detect attacks including DoS, DDoS, and Probe.
- Built a real-time detection pipeline that analyzed live network traffic, extracted features, performed ML inference, and classified malicious activities with automated threat detection.
- Designed an interactive CLI-based monitoring dashboard for real-time observability, providing live traffic monitoring, attack detection, historical attack logs, blocked IP tracking, and system status.
- Implemented a malware detection module to analyze suspicious files and identify potentially malicious content alongside network-based threat detection.
- Developed an automated IP blocking mechanism that isolated malicious sources upon attack detection while maintaining detailed logs for auditing and incident analysis.
- Exposed the trained ML model through a Flask REST API, enabling real-time predictions and integration with external applications.
- Performed model evaluation, debugging, and feature validation to improve detection accuracy and reduce false positives.

ReWear — Clothing Exchange Platform

- Built a full-stack web application (Laravel/PHP backend) with an admin dashboard for managing user accounts, product listings, transaction states, and inventory verification.
- Designed an asynchronous ML-based image validation pipeline to automatically detect and flag defective product conditions before catalog sync, reducing manual review overhead.
- Implemented structured exception handling and backend log monitoring to surface pipeline failures early, supporting faster debugging during QA and production runs.

X-Ray Diagnostic Classifier

- Developed a Convolutional Neural Network (CNN) using TensorFlow and Keras to classify chest X-ray images into Normal, COVID-19, Viral Pneumonia, and Lung Opacity categories.
- Trained and evaluated the model on labeled chest X-ray image datasets by performing image preprocessing, normalization, and data augmentation to improve model generalization and classification accuracy.
- Optimized model performance through iterative experimentation, hyperparameter tuning, and evaluation using accuracy, precision, recall, and confusion matrix analysis.
- Developed an AI-based diagnostic classification solution aimed at supporting early disease screening.

CERTIFICATIONS

- Google Technical Support Fundamentals — Google
- AWS Academy Cloud Foundations — Amazon Web Services
- AI and Machine Learning on Google Cloud — Google Cloud
- AI for Medicine — Coursera
- AI for Business: Generation and Prediction — Coursera
- Google Cybersecurity — Coursera
- Generative AI: Prompt Engineering Basics — Coursera